基于 SGLangJax 的 mimo_v2_flash 报告

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测试说明：BS=64, 非 MTP，decode

- 性能
  - media ITL：20.86ms > 29.40ms * 2/3 ~= 19.6ms （baseline vs SGLangJax，后者快 6%）
  - peak output token throughput: 3490 * 2/3 ~= 2327 < 2560 （baseline vs SGLangJax，后者高 10%）
  - 备注：baseline 8 卡，SGLangJax 16 卡（单卡 32GB）
- 精度
  - 和 HuggingFace （GSM8K=92.3）对比，SGLangJax 92.27，基本一致。

前言

- 软件说明
  - Jax 基础版本

```
pip list | egrep 'jax|flax|libtpu'flax                    0.12.4
jax                     0.8.1
jaxlib                  0.8.1
jaxtyping               0.3.9
libtpu                  0.0.30
sglang-jax              0.0.2          /sglang-jax/python
```

  - SGLangJAX:
    - branch: epic/mimo-v2-flash
    - commit: cef4a181508ef22b452a38fe4210478e2e3672b1
  - 评测相关
    - evalscope：0.17.1
- 硬件说明
  - tpu-v6e
  - topo：4x4
- 启动以及 benchmark 命令相关参考 [MiMo-V2-Flash性能测试](#) 。

注意：下述测试相比上述文档没有开启 radix cache，我们本次主要关注非 MTP 下的 decode 吞吐。

性能测试

命令相关

```
# launch server
export RANK=0 && JAX_COMPILATION_CACHE_DIR=/tmp/jit_cache python -u -m sgl_jax.launch_server \
    --model-path /models/MiMo-V2-Flash \
    --trust-remote-code \
    --tp-size 16 --dp-size 4 --ep-size 16 \
    --moe-backend epmoe \
    --nnodes 4 --node-rank "$RANK" \
    --dist-init-addr 10.31.186.7:30000 \
    --host 0.0.0.0 --port 30271 \
    --page-size 256 \
    --context-length 262144 \
    --disable-radix-cache \
    --chunked-prefill-size 2048 \
    --dtype bfloat16 \
    --mem-fraction-static 0.95 \
    --swa-full-tokens-ratio 0.2 \
    --skip-server-warmup \
    --log-level info \
    --max-running-requests 128
```

```
# benchmark
python3 -m sgl_jax.bench_serving --backend sgl-jax --host 127.0.0.1 --port 30271 --dataset-name random --random-input-len 16384 --random-output-len 1024 --random-range-ratio 1.0 --flush-cache --seed 12345 --request-rate 100 --num-prompts 256 --max-concurrency 64
```

结果记录

- batch_size: 64
- media itl: 29.40ms
- peak output token throughput: 2560

BS=64, Media ITL=29.4ms

复制代码

```
============ Serving Benchmark Result ============
Backend:                                 sgl-jax
Traffic request rate:                    100.0
Max request concurrency:                 64
Successful requests:                     256
Benchmark duration (s):                  400.30
Total input tokens:                      4194304
Total input text tokens:                 4194304
Total generated tokens:                  262144
Total generated tokens (retokenized):    261997
Request throughput (req/s):              0.64
Input token throughput (tok/s):          10477.80
Output token throughput (tok/s):         654.86
Peak output token throughput (tok/s):    2560.00
Peak concurrent requests:                128
Total token throughput (tok/s):          11132.66
Concurrency:                             63.93
----------------End-to-End Latency----------------
Mean E2E Latency (ms):                   99967.13
Median E2E Latency (ms):                 102124.65
P90 E2E Latency (ms):                    102182.53
P99 E2E Latency (ms):                    102191.65
---------------Time to First Token----------------
Mean TTFT (ms):                          34062.26
Median TTFT (ms):                        35028.21
P99 TTFT (ms):                           68888.62
-----Time per Output Token (excl. 1st token)------
Mean TPOT (ms):                          64.42
Median TPOT (ms):                        65.16
P99 TPOT (ms):                           97.95
---------------Inter-Token Latency----------------
Mean ITL (ms):                           64.42
Median ITL (ms):                         29.40
P95 ITL (ms):                            31.94
P99 ITL (ms):                            32.82
Max ITL (ms):                            67344.97
==================================================
```

epic_mimo-v2-flash_cef4a18_benchmark.log
185.8KB

```
log cat epic_mimo-v2-flash_cef4a18_benchmark.log| grep "Decode" | tail -n 10
[2026-04-10 08:26:30 NP0] Decode batch. #running-req: 64, #full token: 1097728, full token usage: 0.92, #swa token: 32768, swa token usage: 0.14, #running-req per DP: [17, 17, 15, 15], gen throughput (token/s): 2184.94, #queue-req: 0,
[2026-04-10 08:26:32 NP0] Decode batch. #running-req: 64, #full token: 1097728, full token usage: 0.92, #swa token: 16384, swa token usage: 0.07, #running-req per DP: [17, 17, 15, 15], gen throughput (token/s): 2179.89, #queue-req: 0,
[2026-04-10 08:26:33 NP0] Decode batch. #running-req: 64, #full token: 1097728, full token usage: 0.92, #swa token: 16384, swa token usage: 0.07, #running-req per DP: [17, 17, 15, 15], gen throughput (token/s): 2184.30, #queue-req: 0,
[2026-04-10 08:26:34 NP0] Decode batch. #running-req: 64, #full token: 1097728, full token usage: 0.92, #swa token: 16384, swa token usage: 0.07, #running-req per DP: [17, 17, 15, 15], gen throughput (token/s): 2179.02, #queue-req: 0,
[2026-04-10 08:26:35 NP0] Decode batch. #running-req: 64, #full token: 1114112, full token usage: 0.94, #swa token: 32768, swa token usage: 0.14, #running-req per DP: [17, 17, 15, 15], gen throughput (token/s): 2182.41, #queue-req: 0,
[2026-04-10 08:26:36 NP0] Decode batch. #running-req: 64, #full token: 1114112, full token usage: 0.94, #swa token: 32768, swa token usage: 0.14, #running-req per DP: [17, 17, 15, 15], gen throughput (token/s): 2179.76, #queue-req: 0,
[2026-04-10 08:26:37 NP0] Decode batch. #running-req: 64, #full token: 1114112, full token usage: 0.94, #swa token: 32768, swa token usage: 0.14, #running-req per DP: [17, 17, 15, 15], gen throughput (token/s): 2178.12, #queue-req: 0,
[2026-04-10 08:26:39 NP0] Decode batch. #running-req: 64, #full token: 1114112, full token usage: 0.94, #swa token: 16384, swa token usage: 0.07, #running-req per DP: [17, 17, 15, 15], gen throughput (token/s): 2175.62, #queue-req: 0,
[2026-04-10 08:26:40 NP0] Decode batch. #running-req: 64, #full token: 1114112, full token usage: 0.94, #swa token: 16384, swa token usage: 0.07, #running-req per DP: [17, 17, 15, 15], gen throughput (token/s): 2170.79, #queue-req: 0,
[2026-04-10 08:26:41 NP0] Decode batch. #running-req: 64, #full token: 1114112, full token usage: 0.94, #swa token: 16384, swa token usage: 0.07, #running-req per DP: [17, 17, 15, 15], gen throughput (token/s): 2172.54, #queue-req: 0,
```

精度测试

命令相关

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```
# launch server
export RANK=0 && JAX_COMPILATION_CACHE_DIR=/tmp/jit_cache python -u -m sgl_jax.launch_server \
    --model-path /models/MiMo-V2-Flash \
    --trust-remote-code \
    --tp-size 16 --dp-size 4 --ep-size 16 \
    --moe-backend epmoe \
    --nnodes 4 --node-rank "$RANK" \
    --dist-init-addr 10.31.147.7:30000 \
    --host 0.0.0.0 --port 30271 \
    --page-size 256 \
    --context-length 262144 \
    --disable-radix-cache \
    --chunked-prefill-size 2048 \
    --dtype bfloat16 \
    --mem-fraction-static 0.95 \
    --swa-full-tokens-ratio 0.2 \
    --skip-server-warmup \
    --log-level info \
    --max-running-requests 128

# evalscope==0.17.1, 采样参数参考 https://huggingface.co/XiaomiMiMo/MiMo-V2-Flash
evalscope eval  \
--model XiaomiMiMo/MiMo-V2-Flash \
--api-url http://127.0.0.1:30271/v1/chat/completions \
--api-key EMPTY \
--eval-type service \
--datasets gsm8k \
--eval-batch-size 32 \
--generation-config '{"temperature": 0.8,"top_p":0.95}'
```

结果记录

三次均值：0.9227。

复制代码

```
# 第一次
+---------------+-----------+-----------------+----------+-------+---------+---------+
| Model         | Dataset   | Metric          | Subset   |   Num |   Score | Cat.0   |
+===============+===========+=================+==========+=======+=========+=========+
```

```
| MiMo-V2-Flash | gsm8k     | AverageAccuracy | main     |  1319 |  0.9105 | default |
+---------------+-----------+-----------------+----------+-------+---------+---------+

# 第二次
+---------------+-----------+-----------------+----------+-------+---------+---------+
| Model         | Dataset   | Metric          | Subset   |   Num |   Score | Cat.0   |
+===============+===========+=================+==========+=======+=========+=========+
| MiMo-V2-Flash | gsm8k     | AverageAccuracy | main     |  1319 |  0.9219 | default |
+---------------+-----------+-----------------+----------+-------+---------+---------+

# 第三次(避免长尾等太久，加了 "max_tokens":65536)
+---------------+-----------+-----------------+----------+-------+---------+---------+
| Model         | Dataset   | Metric          | Subset   |   Num |   Score | Cat.0   |
+===============+===========+=================+==========+=======+=========+=========+
| MiMo-V2-Flash | gsm8k     | AverageAccuracy | main     |  1319 |  0.9356 | default |
+---------------+-----------+-----------------+----------+-------+---------+---------+
```

结论

- 性能角度：
  - baseline（关注 非 MTP decode， bs =64）：
    - media ITL：20.86ms
    - peak output token throughput: 3490，来自日志
  - SGLangJax（非 MTP，bs=64）
    - media ITL：29.40ms
    - peak output token throughput: 2560
  - 结论：
    - 转换系数：考虑到 baseline 是 8 卡，SGLangJax 用的是 16 卡，目标单卡为 1/3，因此转换系数为2/3
    - ***media ITL：29.40ms * 2/3 ~= 19.6ms > 20.86ms***
    - ***peak output token throughput: 3490 * 2/3 ~= 2327 < 2560***
- 精度角度：和 HuggingFace （GSM8K=92.3）对比，SGLangJax 92.27，基本一致。

未来提升方向

1. 目前16k inputs 1k outputs，并发只能达到64，当前实现极限 86BS
2. attention kernel 目前采用 default block size，还需要 tune
3. fused moe kernel 还没有上，预期还有较大的优化空间，目前是 epmoe
4. decode前预处理操作 琐碎优化点 ： 通过在初始化模型时前置，或者修改不当写法修复。能提升0.14ms